Logistic Regression

In [2]:
```
import pandas as pd
import numpy as np
from sklearn.linear_model import LogisticRegression
import matplotlib.pyplot as plt
import streamlit as st
```

In [165…
```
df=pd.read_csv('Logic.csv')
df
```

Out[165…

	hours_studied	attendance_percent	previous_score	pass_exam
0	3.75	51.6	75.3	1
1	9.51	81.8	44.6	1
2	7.32	65.7	48.9	1
3	5.99	75.4	89.4	1
4	1.56	95.4	73.4	1
...	...	...	...	...
95	4.94	67.5	68.7	1
96	5.23	86.3	82.3	1
97	4.28	94.9	51.9	1
98	0.25	94.4	74.3	0
99	1.08	89.0	44.7	0

100 rows × 4 columns

In [166…
```
model=LogisticRegression()
model.fit(df[['hours_studied','attendance_percent','previous_score']],df['pass_exam'])
```

Out[166…

```
▾ LogisticRegression
LogisticRegression()
```

In [177…

```
Predicted_Values=model.predict(df[['hours_studied','attendance_percent','previous_score']])
df['pass_probability']=Predicted_Values
df
Values= model.score(df[['hours_studied','attendance_percent','previous_score']],df['pass_exam'])
import streamlit as st
st.subheader("Model Score")
Values
```

```
2026-03-18 11:30:24.602 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in ba
re mode.
2026-03-18 11:30:24.603 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in ba
re mode.
```

Out[177…

```
0.99
```

In [158…

```
model.predict([[4.4,93.5,81.9]])
```

```
C:\Users\agraw\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2739: UserWarning: X does not have valid featu
re names, but LogisticRegression was fitted with feature names
  warnings.warn(
```

Out[158…

```
array([1])
```

In [72]:

```
model.predict([[1.5,45.7,64.5]])
```

```
C:\Users\agraw\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2739: UserWarning: X does not have valid featu
re names, but LogisticRegression was fitted with feature names
  warnings.warn(
```

Out[72]:

```
array([0])
```

In [117…

```
model.predict([[0.4,54,74]])
```

```
C:\Users\agraw\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2739: UserWarning: X does not have valid featu
re names, but LogisticRegression was fitted with feature names
  warnings.warn(
```

Out[117…
```
array([0])
```

In [118…
```
model.predict([[4.5,54,74]])
```

```
C:\Users\agraw\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2739: UserWarning: X does not have valid featu
re names, but LogisticRegression was fitted with feature names
  warnings.warn(
```

Out[118…
```
array([1])
```

In [74]:
```
model.predict([[0.5,54,74]])
```

```
C:\Users\agraw\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2739: UserWarning: X does not have valid featu
re names, but LogisticRegression was fitted with feature names
  warnings.warn(
```

Out[74]:
```
array([0])
```

In [119…
```
model.predict([[7.5,54,22.5]])
```

```
C:\Users\agraw\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2739: UserWarning: X does not have valid featu
re names, but LogisticRegression was fitted with feature names
  warnings.warn(
```

Out[119…
```
array([1])
```

In [120…
```
model.score(df[['hours_studied','attendance_percent','previous_score']],df['pass_exam'])
```

Out[120…
```
0.99
```

In [160…
```
Data={'hours_studied':[6.7, 2.1, 5.2, 3.4, 8.4, 0.5, 5.7, 4.5, 3.3, 0.1],
     'attendance_percent':[78.5, 45.4 , 66.5, 32.5, 95.4, 39.6, 45.6, 77.4, 78.6, 33.5],
     'previous_score':[74.6, 57.4, 83.5, 44.5, 94.7, 33.5, 44.5, 63.5, 31.5, 95.5]}

df1=pd.DataFrame(Data)
df1
```

Out[160…

	hours_studied	attendance_percent	previous_score
0	6.7	78.5	74.6
1	2.1	45.4	57.4
2	5.2	66.5	83.5
3	3.4	32.5	44.5
4	8.4	95.4	94.7
5	0.5	39.6	33.5
6	5.7	45.6	44.5
7	4.5	77.4	63.5
8	3.3	78.6	31.5
9	0.1	33.5	95.5

In [161…

```
Predicted_Result= model.predict(df1[['hours_studied','attendance_percent','previous_score']])
# score1=model.score(df1[['hours_studied','attendance_percent','previous_score']],Predicted_Result)
# score1
df1['Predicted Result']= Predicted_Result
```

In [162…

```
model.score(df1[['hours_studied','attendance_percent','previous_score']],Predicted_Result)
```

Out[162…

```
1.0
```

In [175…

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import streamlit as st
# Show predictions
st.title("Students Dashboard")
st.subheader("Predictions")
st.table(df1)
st.title("Model Score")
score=model.score(df1[['hours_studied','attendance_percent','previous_score']],df1['Predicted Result'])
st.write(score)
```

```
#Graph
fig,ax=plt.subplots()
ax.scatter(df1['attendance_percent'],df1['previous_score'])
ax.plot(df1['attendance_percent'],df1['previous_score'])
ax.set_xlabel("Attendence_Percentage")
ax.set_ylabel("Previous_score")

st.pyplot(fig)
```

```
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re mode.
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2026-03-18 11:24:22.740 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in ba
re mode.
2026-03-18 11:24:22.741 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in ba
re mode.
```

Out[175…

```
DeltaGenerator()
```

In [ ]:

```

```